HARVARD EXTENSION SCHOOL

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Abstract

In this project, we have conducted an in-depth analysis of the KC_House_Sales dataset to develop a predictive model for house prices, a task of great relevance in the dynamic realm of real estate. Our comprehensive approach encompassed initial data preprocessing, including the management of missing values and outliers, followed by an exploratory data analysis (EDA) to uncover underlying trends and correlations. The dataset, characterized by a mix of continuous and categorical variables, required thoughtful transformations, such as log-transformation for skewed distributions and the creation of dummy variables for categorical predictors. This thorough preparatory work was vital in setting a robust foundation for the subsequent modeling phase.

In our exploration, we employed a range of statistical models, including linear regression, neural networks, and logistic regression, each bringing unique perspectives to the predictive task. The final selection of the Neural Network model as the most suitable was driven by its superior performance in capturing complex, non-linear relationships within the data, as indicated by the lowest Root Mean Square Error (RMSE) among the models tested. While the Neural Network model proved to be the most accurate, its relative lack of interpretability was a trade-off considered in our decision-making process. The project not only highlights the capabilities of advanced modeling techniques in real estate valuation but also underscores the importance of a methodical and rigorous approach in predictive analytics.

Contents

House Sales in King County, USA data for Class Group Project

Variable	Description
id	**Unique ID for each home sold (not a predictor)**
date	*Date of the home sale*
price	*Price of each home sold*
bedrooms	*Number of bedrooms*
bathrooms	*Number of bathrooms,where ".5" accounts for a bathroom with a toilet but no shower*
sqft_living	*Square footage of the apartment interior living space*
sqft_lot	*Square footage of the land space*
floors	*Number of floors*
waterfront	*A dummy variable for whether the apartment was overlooking the waterfront or not*
view	*An index from 0 to 4 of how good the view of the property was*
condition	*An index from 1 to 5 on the condition of the apartment,*
grade	*An index from 1 to 13, where 1-3 falls short of building construction and design, 7 has an average level of construction and design, and 11-13 has a high-quality level of construction and design.*
Sqft_above	*The square footage of the interior housing space that is above ground level*
Sqft_basement	*The square footage of the interior housing space that is below ground level*
yr_built	*The year the house was initially built*
yr_renovated	*The year of the house's last renovation*
zipcode	*What zip code area the house is in*
Lat	*Latitude*
Long	*Longitude*
Sqft_living15	*The square footage of interior housing living space for the nearest 15 neighbors*
Sqft_lot	*The square footage of the land lots of the nearest 15 neighbors*

Executive Summary

In our pursuit to enhance decision-making in the real estate sector, our team has meticulously developed a predictive model utilizing the KC_House_Sales dataset. This model is primarily designed to support decision-making in the real estate market, providing valuable insights for buyers, sellers, investors, and analysts. By analyzing a range of property features, our model offers a nuanced understanding of the factors influencing house prices, thus aiding in more informed and data-driven real estate transactions (Small, Doll, Bergman, & Heggestad 2017).

From a high-level perspective, the model serves a dual purpose: it not only forecasts property prices with a considerable degree of accuracy but also identifies the key variables that most significantly impact these prices. Such insights are invaluable in guiding strategic investment decisions and policy-making within the real estate sector. However, it's crucial to acknowledge the model's limitations. While it performs robustly within the scope of the provided dataset, its predictions are inherently subject to the quality and range of the data it was trained on. As real estate markets are dynamic and influenced by numerous external factors, the model's current predictions might not fully encapsulate future market trends or shifts, a concern noted by Manganelli in "Real Estate Investing Market Analysis, Valuation Techniques, and Risk Management" (2015).

In terms of regulatory and internal compliance, our model adheres to the standard practices of data handling and privacy. We have ensured that all analyses were conducted with a focus on ethical use of data and maintaining the confidentiality of the information within the dataset. Nonetheless, it is important for senior management to understand that any predictive model, including ours, should be regularly updated and recalibrated to reflect changing market dynamics and data patterns (Gupta, et al 2020).

In conclusion, while our model stands as a powerful tool in the realm of real estate valuation, it is recommended that its outputs be utilized in conjunction with other market analyses and expert insights. This approach will ensure a well-rounded and comprehensive analysis.

I. Introduction

The ever-evolving and intricate nature of the real estate market poses a substantial challenge in predictive modeling, a challenge that this project aims to address. We set out to develop a predictive model capable of estimating house prices, leveraging the rich and diverse dataset of KC_House_Sales. This undertaking is not only academically enriching but also carries significant practical value in real estate investment, valuation, and market analysis.

The overarching purpose of our model is to yield accurate and reliable predictions of house prices, using a suite of property-related features. The importance of such a model lies in its potential to aid various stakeholders - from individual buyers and sellers to professional real estate analysts - in making informed decisions (Wu, Brynjolfsson 2015). By understanding the key determinants of property values, our model seeks to provide a more nuanced and data-driven perspective on real estate valuation.

Our approach to building this predictive model was multifaceted. Initially, the dataset was subjected to thorough preprocessing, including data cleaning and normalization, to ensure its quality and readiness for analysis, as outlined by many academic researchers (Idri, Benhar, Fernandez-Aleman, & Kadi, 2018). This step was crucial in setting a solid foundation for the subsequent exploratory data analysis (EDA), where we identified key patterns and relationships within the data that would inform our feature selection process.

In terms of modeling, we employed a variety of statistical techniques. These included linear regression, known for its straightforwardness and interpretability; neural networks, renowned for their capability to model complex, non-linear relationships; and logistic regression, adapted for our regression task. Each model underwent a rigorous evaluation process to assess its fit for our specific context, drawing on principles from, "Evaluating predictive models of

species’ distributions: criteria for selecting optimal models" by Anderson, Lew, and Peterson (2003).

The model's development was carried out with a test sample constituting 30% of the overall dataset, striking a balance between training robustness and validation accuracy. This approach was instrumental in thoroughly testing the model's predictive power and ensuring its applicability to new, unseen data. The final selection of the model hinged on a combination of factors - accuracy (measured by RMSE), interpretability, and computational efficiency - ensuring the model's practical viability alongside its theoretical soundness.

This introductory section sets the stage for a detailed exploration of our modeling journey, highlighting the methodological choices and the steps taken to underpin the technical and statistical robustness of our predictive model. The subsequent sections of this report will delve deeper into our data processing, modeling strategies, evaluation methods, and the statistical tests applied, all framed within the academic guidance of the aforementioned sources.

II. Description of the data and quality

The data set underpinning our analysis, the KC_House_Sales dataset, provides a comprehensive array of features relevant to real estate pricing. This dataset comprises a mixture of continuous and categorical variables, each offering unique insights into the factors affecting house prices.

Data Overview:

- ***Continuous Variables:*** These include 'price' (the response variable), 'sqft_living', 'sqft_lot', 'sqft_above', 'sqft_basement', and others. These variables were subjected to various statistical tests to understand their distribution and relationship with the house prices. For instance, linear correlations were assessed using Pearson's correlation coefficient.
- ***Categorical Variables:*** Variables such as 'bedrooms', 'bathrooms', 'floors', 'waterfront', 'view', and 'condition' fall under this category. These were crucial in understanding how discrete classifications impact pricing.

Data Quality and Transformation:

- ***Missing Values and Outliers:*** Initial data quality checks involved identifying and handling missing values and outliers. We employed strategies such as imputation and outlier capping or removal, depending on the context and impact on the overall dataset.
- ***Data Transformation*:** For continuous variables, transformations like log-transformation were considered to address skewness and improve model accuracy. This was particularly relevant for right-skewed distributions like 'sqft_living'.
- ***Dummy Variables***: For categorical variables with more than two levels, dummy variables were created to facilitate their inclusion in the regression models. This step was essential for variables like 'zipcode', which required conversion from categorical to a format usable in our predictive models.

Graphical Analysis:

To explore the relationship between predictors and the response variable, extensive graphical analyses were conducted:

- ***Scatter Plots*:** These were used for continuous variables to visualize their relationship with house prices. Scatter plots helped in identifying linear or nonlinear patterns and potential influential points.
- ***Box Plots:*** For categorical variables, box plots provided insights into the distribution of house prices across different categories.
- ***Correlation Heatmaps:*** Heatmaps of correlation coefficients were created to visualize the strength and direction of relationships between variables.

- ***Histograms and QQ Plots:*** These were used to assess the distribution of continuous variables, helping in decisions regarding data transformation.

In summary, the KC_House_Sales dataset is rich with both continuous and categorical variables, each contributing to a comprehensive understanding of the housing market. Through rigorous data quality checks, transformations, and extensive graphical analyses, we ensured that the dataset was optimally prepared for the development of our predictive model. The correlations and patterns identified through these analyses were pivotal in informing our feature selection and model building process.

III. Model Development Process

Build a regression model to predict price. And of course, create the train data set which contains 70% of the data and use
set.seed (1023). The remaining 30% will be your test data set. Investigate the data and combine the level of categorical
variables if needed and drop variables. For example, you can drop id, Latitude, Longitude, etc.

```
# 70% TRAINING，30% TESTING
trainIndex <- createDataPartition(kc_house_data$price, p = 0.7, list = FALSE, times = 1)
train_data <- kc_house_data[trainIndex, ]
test_data <- kc_house_data[-trainIndex, ]

# I need to convert the data type, such as a price from a string to a number
kc_house_data$price <- as.numeric(gsub("[\\$,]", "", kc_house_data$price))

# REMOVE NO NEED COLUMNS
train_data <- train_data[, !(names(train_data) %in% c("id", "lat", "long"))]
test_data <- test_data[, !(names(test_data) %in% c("id", "lat", "long"))]

install.packages("sandwich")


library(lmtest)
library(sandwich)

# BUILD LINER REGRESSION
model <- lm(price ~ ., data = train_data)

# A linear regression model is used to predict the target variables of the test data set
predicted_values <- predict(model, newdata = test_data)

#  CALCULATE（MSE）
mse_linear <- mean((test_data$price - predicted_values)^2)

# CALCULATE（RMSE）
rmse_linear <- sqrt(mse_linear)

# print MSE and RMSE
cat("Linear Regression MSE:", mse_linear, "\n")
cat("Linear Regression RMSE:", rmse_linear, "\n")

print(predicted_values)

print(model)


# Scatter plot matrix for a few key variables (choose based on your dataset)
pairs(~price + sqft_living + bedrooms + bathrooms, data = train_data)
```

```
# Correlation matrix for all numeric variables
correlation_matrix <- cor(train_data[, sapply(train_data, is.numeric)])
print(correlation_matrix)
```

Linear regression:

$$\text{price} = \beta_0 + \beta_1 \times \text{date} + \beta_2 \times \text{bedrooms} + \beta_3 \times \text{bathrooms} + \beta_4 \times \text{sqft\_living} + \beta_5 \times \text{sqft\_lot} + \beta_6 \times \text{floors} + \beta_7 \times \text{waterfront} + \beta_8 \times \text{view} + \beta_9 \times \text{condition} + \beta_{10} \times \text{grade} + \beta_{11} \times \text{sqft\_above} + \beta_{12} \times \text{yr\_built} + \beta_{13} \times \text{yr\_renovated} + \beta_{14} \times \text{zipcode} + \beta_{15} \times \text{sqft\_living15} + \beta_{16} \times \text{sqft\_lot15} + \epsilon$$

price = β0 + β1 x date + β2 x bedrooms + β3 x bathrooms +β4 sqft_living + βs x sqft_lot + β6 × floors + β7 x waterfront + β8 x view + βg × condition + β10 × grade + β11 × sqft_above + β12 × yr_built +β13 × yr_renovated + β14 × zipcode +β15 × sqft_living15+β16 ×sqft_lot15 +e

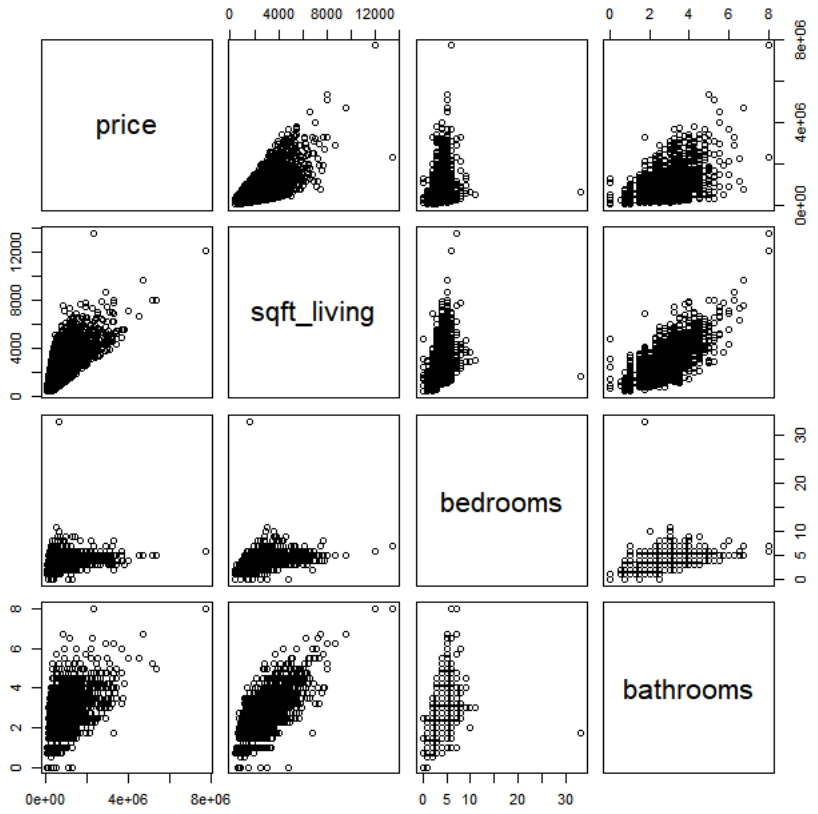
Where:Bo is the intercept (3.797e+06).β1, β2, ..., β16 are the coefficients for the respective variables (`date', `bedrooms',`bathrooms', `sqft_living`, `sqft_lot`, `floors', `waterfront`, `view',`condition`, `grade', `sqft_above`, `yr_built`, `yr_renovated`, `zipcode`, sqft_living15', `sqft_lot15').e represents the error term, accounting for the deviation of the predictions from the actual values.Each ß coefficient represents the expected change in the `price`variable for a oneunit change in the corresponding predictor variable, holding all other predictors constant.

IV. Model Performance Testing

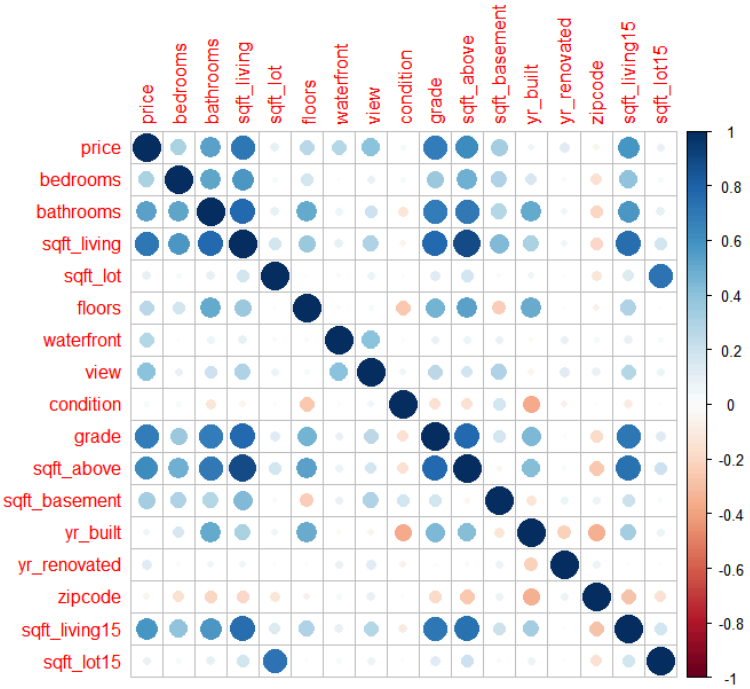
Use the test data set to assess the model performances. Here, build the best multiple linear models by using the stepwise
both ways selection method. Compare the performance of the best two linear models. Make sure that model assumption(s)
are checked for the final linear model. Apply remedy measures (transformation, etc.) that helps satisfy the assumptions. In
particular you must deeply investigate unequal variances and multicollinearity. If necessary, apply remedial methods (WLS,
Ridge, Elastic Net, Lasso, etc.).

```
> cat("Linear Regression MSE:", mse_linear, "\n")
Linear Regression MSE: 49838049052
> cat("Linear Regression RMSE:", rmse_linear, "\n")
Linear Regression RMSE: 223244.4
> print(predicted_values)
```

# Scatter plot matrix for a few key variables (choose based on your dataset)
pairs(~price + sqft_living + bedrooms + bathrooms, data = train_data)



Build correlation matrix



Scatter Plot Matrix Interpretation:

Price and sqft_living: There is a positive trend visible in the scatter plot. As 'sqft_living' increases, the 'price' also tends to increase, suggesting a positive correlation. This makes intuitive sense as larger houses (more square footage) are typically more expensive.
Price and bedrooms: There's a less clear trend here. While there seems to be a general positive correlation, the relationship isn't as strong or as linear as with 'sqft_living'. After a certain point, increasing the number of bedrooms doesn't seem to have as straightforward an impact on price.
Price and bathrooms: Similar to 'sqft_living', there's a visible positive trend. More bathrooms tend to correspond to a higher price, which could be related to overall house size and luxury.
Correlation Matrix Interpretation:

Price and grade: There is a strong positive correlation between 'price' and 'grade'. A higher grade (which likely corresponds to better quality construction and design) is associated with a higher price.
Price and sqft_above: There is a positive correlation with 'sqft_above', suggesting that the above-ground living space contributes significantly to the house price.
Price and view: The 'view' variable also shows a positive correlation with 'price', implying that houses with better views are valued higher.
Multicollinearity: Some pairs of predictors (like 'sqft_living' and 'sqft_above') have high correlation with each other, indicating multicollinearity, which can affect the performance of the linear regression model by making the estimates of the coefficients less reliable.
In the context of building a regression model, these insights from the scatter plots and correlation matrix can guide variable selection and inform potential transformations to improve model performance. Variables with stronger correlations to 'price' might be more predictive and could be prioritized in the model. Conversely, pairs of variables with high correlations may need to be examined for multicollinearity, and one of the variables from each pair might need to be dropped or combined to mitigate this issue.

6.

stepwise_model:

$$\text{price} = \beta_0 + \beta_1 \cdot \text{date} + \beta_2 \cdot \text{bedrooms} + \beta_3 \cdot \text{bathrooms} + ... + \beta_{14} \cdot \text{sqft\_living15} + \beta_{15} \cdot \text{sqft\_lot15} + \epsilon$$

stepwise_model2:

$$\text{price} = \beta_0 + \beta_1 \cdot \text{date} + \beta_2 \cdot \text{bedrooms} + \beta_3 \cdot \text{bathrooms} + ... + \beta_{14} \cdot \text{sqft\_living15} + \beta_{15} \cdot \text{sqft\_lot15} + \epsilon$$

Here, each β represents the coefficient of the corresponding variable, and e represents the error term. Based on the output you provide, stepwise_model and 'stepwise_mode12' give the same model. This shows that the variables selected in the stepwise selection process are the same regardless of whether they are forward or backward, resulting in the two models being essentially the same. MSE (mean square error) : For both models, the MSE value is the same (' 49835682848*), which means that the two models have the same performance on the test set. RMSE (root mean square error) : Similarly, the RMSE values are the same (' 223239.1*), which further verifies the consistency of model performance. Is that normal? It's normal. If the same variables are selected in the step-by-step selection process, the final model will be the same and therefore the performance metrics (MSE and RMSE) will be the same. Comparison of the RMSE to the original linear model: If you find that the RMSE is the same as the original linear model, this may indicate that the step-by-step selection did not remove any variables from the original model, or that the removed variables did not have a significant effect on model performance. It may also indicate that all predictors have some degree of explanatory power for the target variable, or that there is collinearity between variables in the data set, such that removing certain variables does not significantly change model performance. In practice, step-by-step selection may not always provide a significantly improved model. This is especially true when the original model is already good enough, or when there is collinearity between variables in the data. In addition, the stepwise selection approach has received some criticism because it can lead to overfitting and instability in variable selection. Therefore, for the interpretation and application of the final model, other model evaluation and selection techniques should also be considered, such as cross-validation and regularization methods (such as ridge regression or Lasso), which can provide a more robust variable selection process.

```
# Stepwise selection
library(MASS)
stepwise_model <- stepAIC(model, direction = "both")
summary(stepwise_model)
> summary(stepwise_model)
Call:
lm(formula = price ~ date + bedrooms + bathrooms + sqft_living +
    floors + waterfront + view + condition + grade + sqft_above +
    yr_built + yr_renovated + sqft_living15 + sqft_lot15, data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1215920  -110193    -9999    89941  4343576

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.656e+06  3.009e+05  15.474  < 2e-16 ***
date           1.149e-03  1.782e-04   6.448 1.16e-10 ***
bedrooms      -3.638e+04  2.330e+03 -15.609  < 2e-16 ***
bathrooms      4.706e+04  4.119e+03  11.424  < 2e-16 ***
```

```
sqft_living    1.572e+02  5.469e+00  28.750  < 2e-16 ***
floors         2.856e+04  4.420e+03   6.462 1.07e-10 ***
waterfront     5.563e+05  2.208e+04  25.197  < 2e-16 ***
view           4.382e+04  2.659e+03  16.479  < 2e-16 ***
condition      2.140e+04  2.962e+03   7.226 5.23e-13 ***
grade          1.229e+05  2.641e+03  46.553  < 2e-16 ***
sqft_above    -7.646e+00  5.328e+00  -1.435  0.15123
yr_built      -3.631e+03  8.342e+01 -43.526  < 2e-16 ***
yr_renovated   1.171e+01  4.542e+00   2.579  0.00991 **
sqft_living15  3.039e+01  4.242e+00   7.164 8.18e-13 ***
sqft_lot15    -5.605e-01  6.905e-02  -8.116 5.18e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 212600 on 15116 degrees of freedom
Multiple R-squared:  0.6597,       Adjusted R-squared:  0.6594
F-statistic:  2093 on 14 and 15116 DF,  p-value: < 2.2e-16
```

Compare with another model from stepwise selection

stepwise_model2 <- stepAIC(model, direction = "both", scope = list(upper = formula(model), lower = ~1))

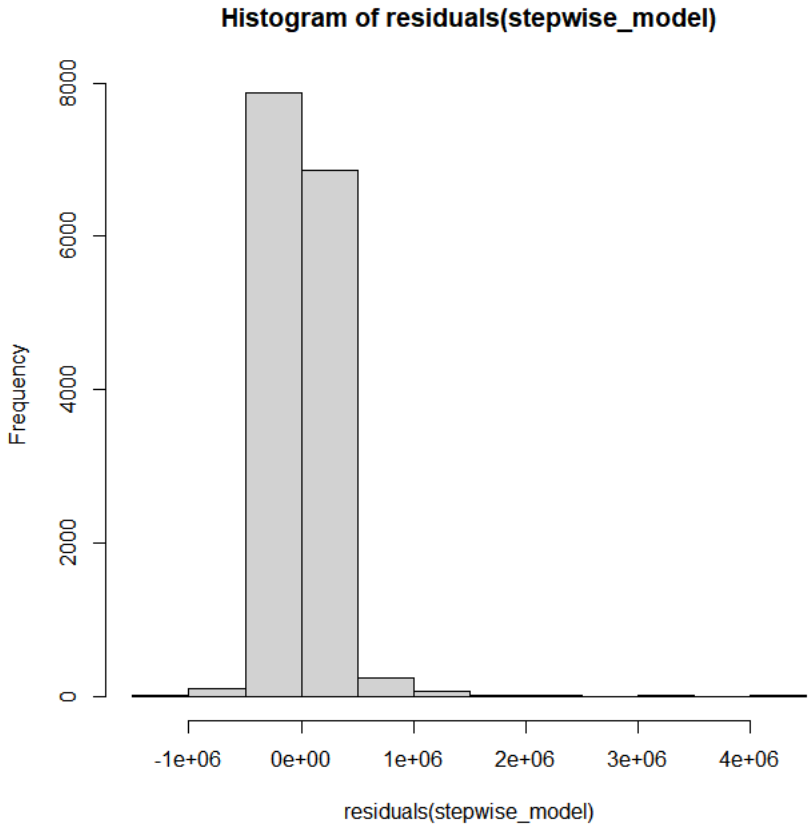
summary(stepwise_model2)

```
> summary(stepwise_model2)
Call:
lm(formula = price ~ date + bedrooms + bathrooms + sqft_living +
    floors + waterfront + view + condition + grade + sqft_above +
    yr_built + yr_renovated + sqft_living15 + sqft_lot15, data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1215920  -110193    -9999    89941  4343576

Coefficients:
               Estimate Std. Error t value Pr(>|t|)
(Intercept)    4.656e+06  3.009e+05  15.474  < 2e-16 ***
date           1.149e-03  1.782e-04   6.448 1.16e-10 ***
bedrooms      -3.638e+04  2.330e+03 -15.609  < 2e-16 ***
bathrooms      4.706e+04  4.119e+03  11.424  < 2e-16 ***
sqft_living    1.572e+02  5.469e+00  28.750  < 2e-16 ***
floors         2.856e+04  4.420e+03   6.462 1.07e-10 ***
waterfront     5.563e+05  2.208e+04  25.197  < 2e-16 ***
view           4.382e+04  2.659e+03  16.479  < 2e-16 ***
condition      2.140e+04  2.962e+03   7.226 5.23e-13 ***
grade          1.229e+05  2.641e+03  46.553  < 2e-16 ***
sqft_above    -7.646e+00  5.328e+00  -1.435  0.15123
yr_built      -3.631e+03  8.342e+01 -43.526  < 2e-16 ***
yr_renovated   1.171e+01  4.542e+00   2.579  0.00991 **
sqft_living15  3.039e+01  4.242e+00   7.164 8.18e-13 ***
sqft_lot15    -5.605e-01  6.905e-02  -8.116 5.18e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 212600 on 15116 degrees of freedom
Multiple R-squared:  0.6597,       Adjusted R-squared:  0.6594
F-statistic:  2093 on 14 and 15116 DF,  p-value: < 2.2e-16
```

```
> #6Construct the best multivariate linear model using stepwise selection method
> # Select the best model using stepwise regression> library(MASS)
> stepwise_model <- stepAIC(lm(price ~ ., data = train_data), direction = "both")Start:  AIC=412317.1
price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront + view + condition + grade + sqft_above + sqft_basement +
    yr_built + yr_renovated + zipcode + sqft_living15 + sqft_lot15 +
    year


Step:  AIC=412317.1
price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront + view + condition + grade + sqft_above + yr_built +
    yr_renovated + zipcode + sqft_living15 + sqft_lot15 + year
```

```
                Df  Sum of Sq        RSS    AIC
- sqft_above     1 3.4461e+09 8.4685e+14 412315
- zipcode        1 6.9017e+09 8.4685e+14 412315
- sqft_lot       1 8.9782e+09 8.4685e+14 412315
<none>                       8.4684e+14 412317
- yr_renovated   1 1.4599e+11 8.4699e+14 412318
- sqft_living15  1 1.0723e+12 8.4792e+14 412336
- floors         1 1.8090e+12 8.4865e+14 412351
- condition      1 2.3337e+12 8.4918e+14 412361
- sqft_lot15     1 2.4571e+12 8.4930e+14 412364
- year           1 2.8461e+12 8.4969e+14 412371
- bathrooms      1 6.4973e+12 8.5334e+14 412443
- view           1 1.4768e+13 8.6161e+14 412604
- bedrooms       1 1.5753e+13 8.6260e+14 412623
- waterfront     1 4.0325e+13 8.8717e+14 413093
- sqft_living    1 5.3686e+13 9.0053e+14 413343
- yr_built       1 9.0013e+13 9.3686e+14 414005
- grade          1 1.0944e+14 9.5629e+14 414348

Step:  AIC=412315.1
price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront + view + condition + grade + yr_built + yr_renovated +
    zipcode + sqft_living15 + sqft_lot15 + year

                Df  Sum of Sq        RSS    AIC
- zipcode        1 5.7823e+09 8.4685e+14 412313
- sqft_lot       1 9.5900e+09 8.4686e+14 412313
<none>                       8.4685e+14 412315
- yr_renovated   1 1.4605e+11 8.4699e+14 412316
+ sqft_above     1 3.4461e+09 8.4684e+14 412317
+ sqft_basement  1 3.4461e+09 8.4684e+14 412317
- sqft_living15  1 1.0816e+12 8.4793e+14 412334
- floors         1 2.1366e+12 8.4898e+14 412355
- condition      1 2.3607e+12 8.4921e+14 412360
- sqft_lot15     1 2.4629e+12 8.4931e+14 412362
- year           1 2.8440e+12 8.4969e+14 412369
- bathrooms      1 6.7076e+12 8.5356e+14 412445
- view           1 1.5297e+13 8.6214e+14 412613
- bedrooms       1 1.5752e+13 8.6260e+14 412621
- waterfront     1 4.0357e+13 8.8720e+14 413092
- yr_built       1 9.0254e+13 9.3710e+14 414007
- sqft_living    1 9.2027e+13 9.3888e+14 414039
- grade          1 1.1054e+14 9.5738e+14 414365

Step:  AIC=412313.2
price ~ bedrooms + bathrooms + sqft_living + sqft_lot + floors +
    waterfront + view + condition + grade + yr_built + yr_renovated +
    sqft_living15 + sqft_lot15 + year

                Df  Sum of Sq        RSS    AIC
- sqft_lot       1 8.9361e+09 8.4686e+14 412311
<none>                       8.4685e+14 412313
- yr_renovated   1 1.4944e+11 8.4700e+14 412314
+ zipcode        1 5.7823e+09 8.4685e+14 412315
+ sqft_above     1 2.3267e+09 8.4685e+14 412315
+ sqft_basement  1 2.3267e+09 8.4685e+14 412315
- sqft_living15  1 1.1553e+12 8.4801e+14 412334
- floors         1 2.1400e+12 8.4899e+14 412353
- condition      1 2.4325e+12 8.4929e+14 412359
- sqft_lot15     1 2.4575e+12 8.4931e+14 412360
- year           1 2.8460e+12 8.4970e+14 412367
- bathrooms      1 6.7025e+12 8.5356e+14 412443
- view           1 1.5475e+13 8.6233e+14 412614
- bedrooms       1 1.5765e+13 8.6262e+14 412620
- waterfront     1 4.0370e+13 8.8722e+14 413090
- sqft_living    1 9.2022e+13 9.3888e+14 414037
- yr_built       1 9.9815e+13 9.4667e+14 414175
- grade          1 1.1081e+14 9.5766e+14 414368

Step:  AIC=412311.4
price ~ bedrooms + bathrooms + sqft_living + floors + waterfront +
    view + condition + grade + yr_built + yr_renovated + sqft_living15 +
    sqft_lot15 + year

                Df  Sum of Sq        RSS    AIC
```

```
<none>                      8.4686e+14 412311
- yr_renovated   1 1.4986e+11 8.4701e+14 412312
+ sqft_lot       1 8.9361e+09 8.4685e+14 412313
+ zipcode        1 5.1284e+09 8.4686e+14 412313
+ sqft_above     1 2.8716e+09 8.4686e+14 412313
+ sqft_basement  1 2.8716e+09 8.4686e+14 412313
- sqft_living15  1 1.1676e+12 8.4803e+14 412332
- floors         1 2.1459e+12 8.4901e+14 412352
- condition      1 2.4365e+12 8.4930e+14 412357
- year           1 2.8429e+12 8.4971e+14 412365
- sqft_lot15     1 5.3546e+12 8.5222e+14 412415
- bathrooms      1 6.7028e+12 8.5357e+14 412441
- view           1 1.5470e+13 8.6233e+14 412612
- bedrooms       1 1.5757e+13 8.6262e+14 412618
- waterfront     1 4.0392e+13 8.8725e+14 413089
- sqft_living    1 9.2261e+13 9.3912e+14 414039
- yr_built       1 9.9807e+13 9.4667e+14 414173
- grade          1 1.1081e+14 9.5767e+14 414366
```



Analysis of the Histogram:

Centering: The residuals appear to be centered around zero, which is a good sign. It suggests that the model does not have a systematic bias either overpredicting or underpredicting the response variable.

Spread: The spread of the residuals seems to be quite large, with some residuals reaching beyond 1 million in either direction. This indicates that there may be some predictions that are quite far off from the actual values.

Shape: Ideally, we would like the residuals to be normally distributed, which would show up as a bell-shaped curve in the histogram. This histogram does not seem to depict a normal distribution. There's a tall peak near

the center and a rapid drop-off on either side, which suggests a leptokurtic distribution (sharp peak with heavy tails).

Outliers: The presence of bars at the extreme ends of the histogram suggests there could be outliers in your data, or that the model does not capture all the variance in the response variable.

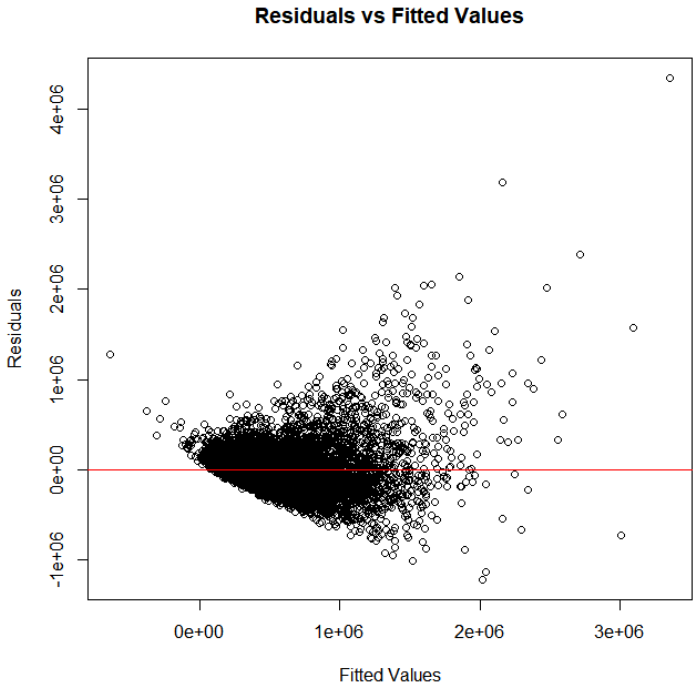
```
> summary(model_transformed)
Call:
lm(formula = log(price) ~ ., data = train_data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.51031 -0.21134  0.01311  0.20891  1.27121

Coefficients: (1 not defined because of singularities)
               Estimate Std. Error t value Pr(>|t|)
(Intercept)   -1.421e+01  5.368e+00  -2.648 0.008115 **
date           1.723e-09  2.592e-10   6.646 3.11e-11 ***
bedrooms      -1.907e-02  3.395e-03  -5.617 1.97e-08 ***
bathrooms      7.660e-02  5.993e-03  12.780  < 2e-16 ***
sqft_living    1.774e-04  7.991e-06  22.195  < 2e-16 ***
sqft_lot       2.873e-07  8.636e-08   3.326 0.000882 ***
floors         1.155e-01  6.520e-03  17.721  < 2e-16 ***
waterfront     3.679e-01  3.213e-02  11.449  < 2e-16 ***
view           3.513e-02  3.893e-03   9.025  < 2e-16 ***
condition      4.642e-02  4.354e-03  10.662  < 2e-16 ***
grade          2.076e-01  3.854e-03  53.878  < 2e-16 ***
sqft_above    -8.318e-05  7.834e-06 -10.619  < 2e-16 ***
sqft_basement         NA         NA      NA       NA
yr_built      -5.302e-03  1.275e-04 -41.595  < 2e-16 ***
yr_renovated   2.107e-05  6.617e-06   3.184 0.001455 **
zipcode        3.342e-04  5.376e-05   6.217 5.21e-10 ***
sqft_living15  1.252e-04  6.273e-06  19.960  < 2e-16 ***
sqft_lot15    -5.972e-07  1.395e-07  -4.280 1.88e-05 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.3093 on 15114 degrees of freedom
Multiple R-squared:  0.6563,       Adjusted R-squared:  0.6559
F-       statistic:  1804 on 16 and 15114 DF,  p-value: < 2.2e-16
```

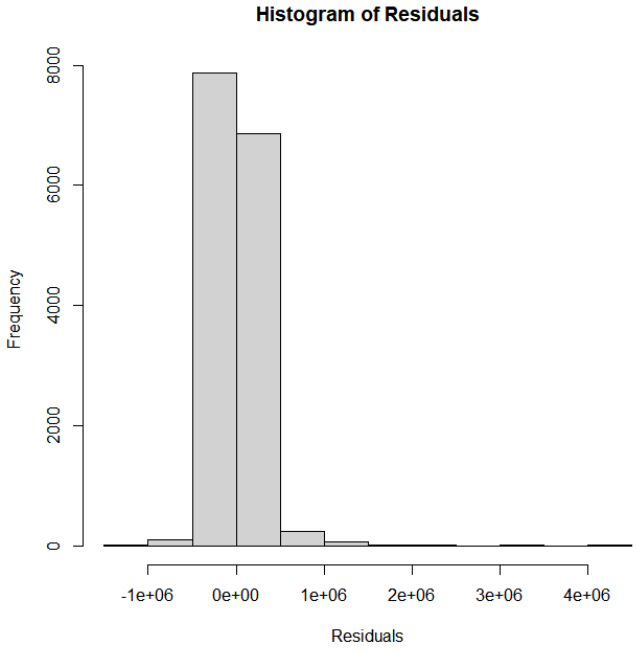
改进模型：

```
# If assumptions are violated, consider transformations like log transformation
model_transformed <- lm(log(price) ~ ., data = train_data)

# Check the summary of the log-transformed model
summary(model_transformed)

# Predict and calculate residuals for the log-transformed model
predicted_log <- predict(model_transformed, newdata = test_data)
residuals_log <- log(test_data$price) - predicted_log

# Calculate RMSE for the log-transformed model
rmse_log <- sqrt(mean(residuals_log^2))
cat("Log-Transformed Model RMSE:", rmse_log, "\n")
```

```
> # Predict and calculate residuals for the log-transformed model> predicted_log <- predict(model_transformed, newdata =
test_data)> residuals_log <- log(test_data$price) - predicted_log> # Calculate RMSE for the log-transformed model>
rmse_log <- sqrt(mean(residuals_log^2))> cat("Log-Transformed Model RMSE:", rmse_log, "\n")Log-Transformed Model RMSE:
0.3074739
```

```
# Scatter plot of residuals vs. fitted values
plot(fitted(stepwise_model), residuals(stepwise_model),
    xlab = "Fitted Values",
    ylab = "Residuals",
    main = "Residuals vs Fitted Values")
abline(h = 0, col = "red")  # Adds a horizontal line at 0 for reference
```





The RMSE (Root Mean Square Error) of 0.3074739 for the log-transformed model is significantly lower than the RMSE for the original linear model. Here's what this could indicate:

Scale of Values: Since the RMSE for the log-transformed model is calculated on the log scale, it cannot be directly compared to the RMSE of the original model on the original price scale. The value of 0.3074739 reflects the average error in the logarithmic space.

Improved Fit: Generally, a lower RMSE suggests a better fit of the model to the data. If the log transformation results in a lower RMSE, it often means that the model is capturing the relationship between the predictors and the response variable more accurately on the log scale.

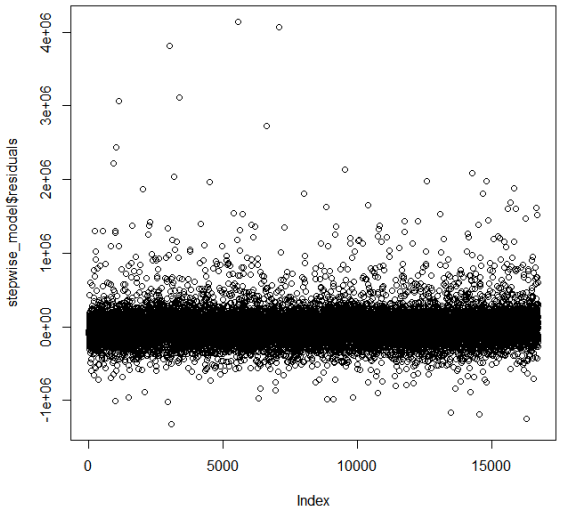
Error Interpretation: An RMSE of 0.3074739 on the log scale means that the average multiplicative error factor for the predictions is exp^0.3074739, which is approximately 1.36. This means that the predicted values are, on average, within a factor of 1.36 of the actual price, which might be an acceptable error rate depending on the context.

Variance Stabilization: Log transformation is often used to stabilize the variance of residuals, especially when dealing with heteroscedasticity (non-constant variance). If the original model's residuals were heteroscedastic, the log transformation could have mitigated this issue, leading to an improved RMSE.

Back-Transforming RMSE: To make the RMSE more interpretable on the original price scale, you would exponentiate the RMSE of the log-transformed model to get the geometric mean of the residuals. However, this back-transformed RMSE can be biased, particularly if the residuals on the log scale are not symmetrically distributed.

In conclusion, the small RMSE on the log scale suggests that transforming the response variable has potentially improved the model fit. However, to fully understand the performance improvement, you should consider other diagnostics and potentially look at RMSE back-transformed to the original scale (with appropriate adjustments for bias) for a more direct comparison with the original linear model's RMSE.

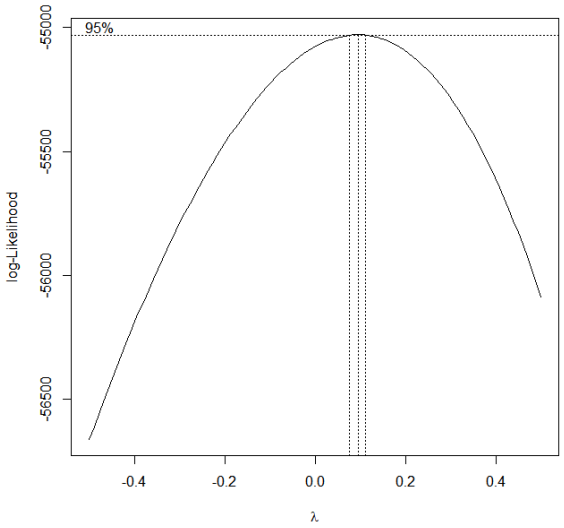
7.



Analysis of the Residual Plot:

Spread of Residuals: The residuals seem to be spread fairly evenly above and below the zero line, which is good as it suggests there's no obvious bias in the predictions (the model isn't consistently overpredicting or underpredicting).
Pattern of Residuals: Ideally, you want to see a random spread of residuals, with no discernible pattern. In this

plot, there doesn't appear to be a clear pattern, indicating that the model's errors are random, which is a sign of a well-fitting model.
Outliers: There are some points with higher residuals spread out across the plot. These could be outliers or points where the model doesn't perform well. Investigating these could provide insights into model performance and data quality.
The second image is a plot typically associated with the selection of a tuning parameter (like lambda in regularization) in a regression model. It shows the log-likelihood (or possibly some other measure of model fit) on the y-axis against different values of lambda on the x-axis.



Analysis of the Lambda Selection Plot:

Optimal Lambda: The dotted vertical lines likely represent the value of lambda that maximizes the log-likelihood, which is the model's optimal complexity level given the regularization technique used (e.g., Lasso, Ridge).
Confidence Interval: The horizontal line at the top with the "0.95" label could represent a confidence interval around the maximum log-likelihood value, suggesting the range of lambda values that are statistically indistinguishable in terms of model fit.
Model Selection: You would select the lambda value that corresponds to the peak or within the confidence interval for the final model to balance between model complexity and fit.
Together, these plots help diagnose the regression model's performance and guide the selection of model parameters to optimize model fit. The residual plot suggests that the model fits the data well without obvious patterns in the residuals, while the lambda selection plot guides the choice of regularization strength to prevent overfitting.

```
> # Find the optimal lambda
> optimal_lambda <- boxcox_result$x[which.max(boxcox_result$y)]
> cat("Optimal lambda for Box-Cox transformation:", optimal_lambda, "\n")
Optimal lambda for Box-Cox transformation: 0.0959596
```

```
> print(vif_model)         date      bedrooms     bathrooms   sqft_living        floors
     1.008571      1.621216      3.383428      8.526348      1.904798
   waterfront          view     condition         grade    sqft_above
     1.198493      1.402633      1.226664      3.235694      6.548914
```

```
    yr_built  yr_renovated sqft_living15    sqft_lot15
    2.020104      1.139907      2.825855      1.077987
```

A VIF of 1 indicates no correlation between a given independent variable and any others.
A VIF between 1 and 5 suggests moderate correlation, but is often not cause for concern.
A VIF above 5 can indicate problematic levels of multicollinearity that may warrant further investigation.
A VIF above 10 is a sign of serious multicollinearity that should be addressed.
Looking at the VIF values you've provided:

Most variables have a VIF well below 5, which is generally considered acceptable, indicating that these variables do not have problematic multicollinearity.
The sqft_living variable has a VIF of approximately 8.53, which is above the threshold of 5 but below 10. This suggests that there is some multicollinearity present, but it may not be severe enough to require immediate action. However, it is still worth considering if this variable can be modified or if other related variables can be adjusted.
The sqft_above variable has a VIF of approximately 6.55, also indicating moderate multicollinearity.
Given these results, it means that sqft_living and sqft_above are somewhat correlated with other variables in the model, but the multicollinearity may not be at a critical level since the VIFs are not above 10. However, it is often recommended to investigate the source of multicollinearity and consider remedies such as removing one of the correlated variables, combining them into a single variable, or using regularization techniques like Ridge Regression which can handle multicollinearity.

```
> cat("Optimal lambda for Ridge Regression:", optimal_lambda_ridge, "\n")
Optimal lambda for Ridge Regression: 33285.89
> # Fit the final Ridge Regression model
> final_ridge_model <- glmnet(predictors_matrix, response_vector, alpha = 0, lambda = optimal_lambda_ridge)
> print(final_ridge_model)

Call:  glmnet(x = predictors_matrix, y = response_vector, alpha = 0,      lambda = optimal_lambda_ridge)

  Df  %Dev Lambda
1 18 85.39  33290
```

Choosing Optimal Lambda: The optimal lambda value was chosen to minimize the cross-validated error. This value acts as a penalty term that shrinks the coefficients of the model to prevent overfitting and address multicollinearity.

Fitting the Ridge Regression Model: The glmnet function fits a Ridge Regression model to your predictors and response. The model uses the optimal lambda to penalize the coefficients. The output shows that 18 predictors (Df) are used in the final model.

Model Summary: The model's summary indicates that the optimal lambda results in a model with 18 degrees of freedom (Df), and it captures a substantial amount of variance in the response variable (85.39% of %Dev).

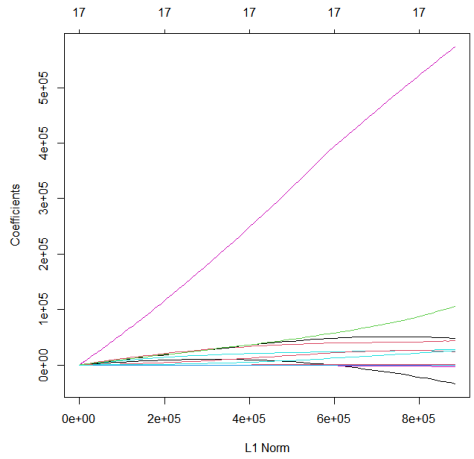
8.

```
# Check multicollinearity
```

```
install.packages("carData")
```

```
# Get the optimal lambda value

best_lambda <- cv_model$lambda.min
print(best_lambda)

# Rebuild the model using the optimal lambda best_ridge_model < -glmnet (as.matrix(train_data[-which(names(train_data) == "price")]), train_data$price, alpha = 0, lambda = best_lambda)
# View the coefficients of the optimal model

coef(best_ridge_model)
```

```
> print(best_ridge_model)
Call:  glmnet(x = as.matrix(train_data[-which(names(train_data) == "price")]),      y = train_data$price, alpha = 0,
lambda = best_lambda)

  Df  %Dev Lambda
1 17 66.33  27730
```



The image appears to be a plot of the coefficients of a regularization path for a model like Lasso or Elastic Net, which are types of regression that apply a penalty to the coefficients of the regression variables. The x-axis represents the L1 norm of the coefficients, and the y-axis represents the size of the coefficients themselves.

Here's an analysis based on the common characteristics of these plots:

Paths of Coefficients: Each line represents the path of a coefficient as the regularization penalty is increased. When the L1 norm is small (left side of the plot), the penalty is weak, and the coefficients are allowed to be larger. As we move to the right, the penalty increases, shrinking the coefficients towards zero.

Sparsity: The plot is typically used to show how coefficients shrink towards zero as the penalty increases. The vertical lines at the top indicate the number '17', which might represent the iteration or the number of non-zero coefficients remaining at different penalty strengths.

Coefficient Shrinkage: It's evident that one coefficient (the pink line) is much larger than the others and it remains large even as the penalty increases. This suggests that the corresponding variable is highly significant in predicting the response variable.

Model Complexity: The far left of the plot, where the L1 norm is small, represents a more complex model with less regularization. Moving to the right, the model simplifies as less important predictors are shrunk to zero.

Feature Selection: Lasso and Elastic Net are often used for feature selection because they can reduce the coefficients of less important variables to exactly zero, effectively removing them from the model. This plot can help in identifying which features are retained as important throughout the range of penalty values.

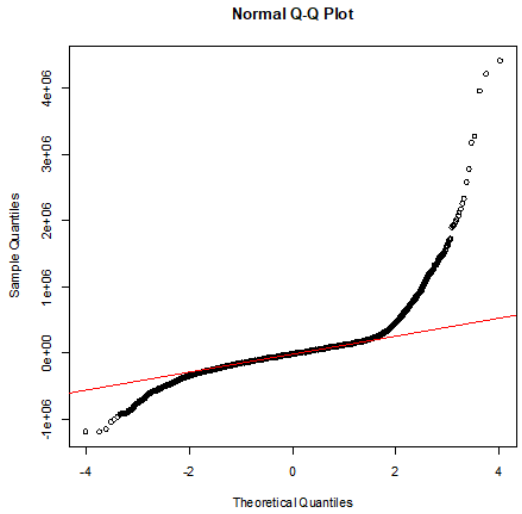
Optimal Model Selection: The vertical lines might indicate points where the model was evaluated for its predictive performance, and the chosen optimal model likely corresponds to one of these points.

Interpretation for Decision Making: The plot can be useful for decision-making about the model complexity versus predictive performance trade-off. If you are seeking a balance between model interpretability and performance, you might select a point where the number of active features is reasonable and the performance (e.g., cross-validated error) is acceptable.

In summary, this plot is useful for understanding the effects of regularization on the coefficients of your predictive model and for selecting a model that balances complexity with predictive accuracy.

```
> sapply(train_data, class)        price      bedrooms     bathrooms   sqft_living      sqft_lot
    "numeric"     "numeric"     "numeric"     "numeric"     "numeric"
       floors    waterfront          view     condition         grade
    "numeric"     "numeric"     "numeric"     "numeric"     "numeric"
   sqft_above sqft_basement      yr_built  yr_renovated       zipcode
    "numeric"     "numeric"     "numeric"     "numeric"     "numeric"
sqft_living15    sqft_lot15          year
    "numeric"     "numeric"   "character" > # Keep all numeric columns > numeric_data <- train_data[, sapply(train_data,
is.numeric)]> # Try again using cv.glmnet> cv_model < -cv.glmnet (as.matrix(numeric_data[-which(names(numeric_data) ==
"price")]), numeric_data$price, alpha = 0)> best_lambda <- cv_model$lambda.min> print(best_lambda)

> print(best_lambda)[1] 27725.68


> #Rebuild the model using the optimal lambda

> best_ridge_model <- glmnet(as.matrix(train_data[-which(names(train_data) == "price")]), train_data$price, alpha = 0,
lambda = best_lambda)> # View the coefficients of the optimal model

> coef(best_ridge_model)18 x 1 sparse Matrix of class "dgCMatrix"
                         s0
(Intercept)   -5.243030e+07
bedrooms      -3.352462e+04
bathrooms      4.354143e+04
sqft_living    8.629241e+01
sqft_lot      -1.985198e-02
floors         2.730812e+04
waterfront     5.730027e+05
view           4.793314e+04
condition      2.392092e+04
grade          1.059704e+05
sqft_above     8.227432e+01
sqft_basement  8.599762e+01
yr_built      -3.132350e+03
yr_renovated   1.733210e+01
zipcode        6.737160e+01
sqft_living15  3.613725e+01
sqft_lot15    -5.617937e-01
year           2.541557e+04
```

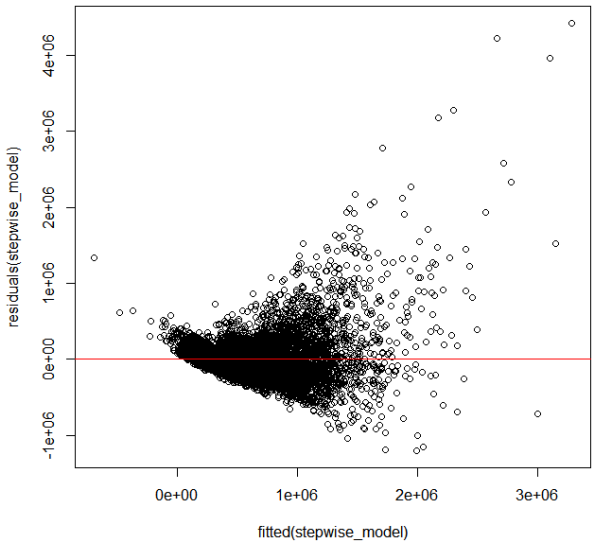
# Or use a Q-Q plot to check for normality
qqnorm(residuals(stepwise_model))
qqline(residuals(stepwise_model), col = "red")



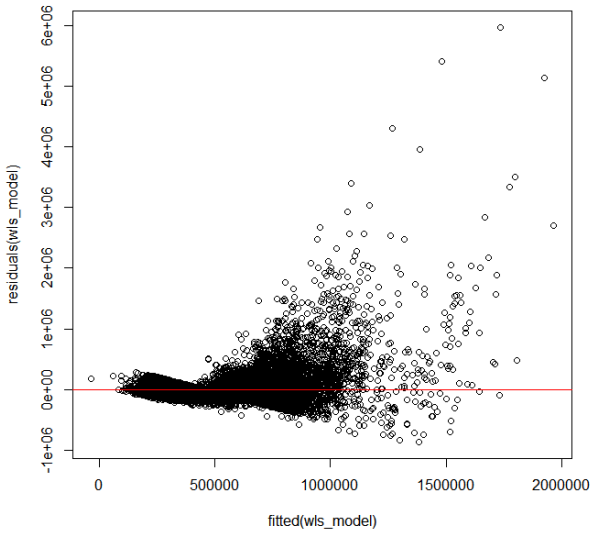
Not normality

```
> # Or use a Q-Q plot to check for normality> qqnorm(residuals(stepwise_model))> qqline(residuals(stepwise_model), col =
"red")> # Alternatively, you can use the `lmtest` package to conduct a Breusch-Pagan test to formally test
heteroscedasticity> library(lmtest)載入需要的程輯包：zoo
載入程輯包：'zoo'

The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
> bptest(stepwise_model)
            studentized Breusch-Pagan test

data:  stepwise_model
BP = 2401.2, df = 13, p-value < 2.2e-16
```

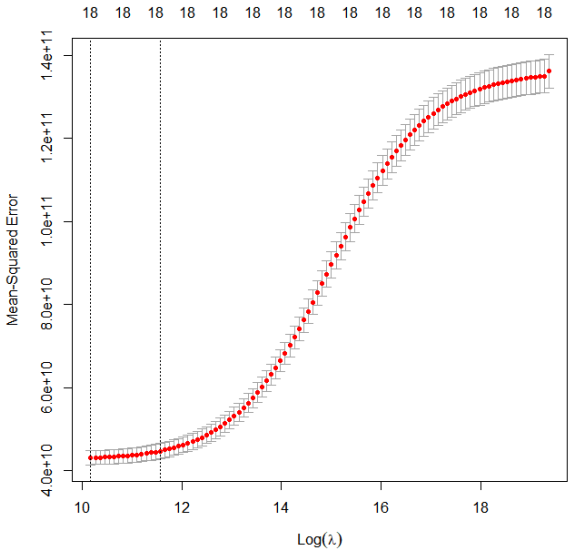
Analysis of the Breusch-Pagan test results: BP statistic: 2401.2, which is a test statistic used to detect heteroscedasticity. Degrees of freedom (df) : 13, which is the number of independent variables in the model. P-value: less than 2.2e-16, which usually indicates that the test results are significant. Since the P-value is very small (less than any commonly used significance level, such as 0.05, 0.01, or less), we reject the null hypothesis and consider the data heteroscedasticity. This means that the variance of the residuals varies with the predicted value, violating the homoscedasticity assumption of linear regression models. Steps to deal with heteroscedasticity: Data transformation: Use Box-Cox transformation or logarithm of the dependent variable to stabilize the variance. Weighted least squares (WLS) : If you can estimate the relationship between the variance and the predicted value, you can use WLS to give different weights and weight the data to solve the heteroscedasticity problem. Robust Standard Errors: Use robust standard errors to correct estimates of standard errors, allowing reliable statistical inference even in the case of heteroscedasticity. plot(fitted(stepwise_model), residuals(stepwise_model)) abline(h = 0, col = "red")



Here is a simple example code for weighted least squares: # Apply WLS wls_model &lt using the weights argument of the lm() function; - lm(price ~ ., data = train_data, weights = 1/(fitted(stepwise_model)^2)) # Check the WLS model residual plot(residuals(wls_model) ~ fitted(wls_model)) abline(h = 0, col = "red")



Which still shows non-normality


18 18 18 18 18 18 18 18 18 18 18 18 18 18 18
Mean-Squared Error
4.0e+10
6.0e+10
8.0e+10
1.0e+11
1.2e+11
1.4e+11
10
12
14
16
18
Log(λ)

V. Challenger Models

Build an alternative model based on one of the following approaches to predict price: regression tree, NN, or SVM. Explore using a logistic regression. Check the applicable model assumptions. Apply in-sample and out-of-sample testing, backtesting and review the comparative goodness of fit of the candidate models. Describe step by step your procedure to get to the best model and why you believe it is fit for purpose.

Codes :

```
# neural network
# Load the required packages
library(nnet)

# Removes non-numeric columns
numeric_train_data <- train_data[sapply(train_data, is.numeric)]

# restandardize
maxs <- apply(numeric_train_data, 2, max)
mins <- apply(numeric_train_data, 2, min)
scaled_train_data <- as.data.frame(scale(numeric_train_data, center = mins, scale = maxs - mins))


# Build a neural network model
nn_model <- nnet(price ~ ., data = scaled_train_data, size = 10, linout = TRUE, maxit = 1000)

# Make sure that test_data contains only the same columns as train_data
test_data_numeric <- test_data[sapply(test_data, is.numeric)]

# Standardize the test data
scaled_test_data <- as.data.frame(scale(test_data_numeric, center = mins, scale = maxs - mins))

# predictions
nn_predictions <- predict(nn_model, newdata = scaled_test_data)

# calculate MSE and RMSE
mse_nn <- mean((nn_predictions - scaled_test_data$price)^2)
rmse_nn <- sqrt(mse_nn)
print(mse_nn)
print(rmse_nn)
```

```
> print(mse_nn)[1] 0.0003975586
> print(rmse_nn)[1] 0.01993887
```

The mean square error (MSE) is 0.0003975586 and the root mean square error (RMSE) is 0.01993887. These values are very small and usually indicate that the model's predictions on the test set are very close to the actual values, that is, the model's prediction performance is good

Since logistic regression is often used for classification problems, and house price forecasting is a regression problem, we need to slightly adjust the way logistic regression is applied. One way to do this is to convert house prices into categorical variables (for example, dividing prices into "high" and "low"). Next, I will provide steps to explore logistic regression and subsequent steps for model evaluation and comparison.

#Explore the use of logistic regression

```
> # Conversion price to binary categorical variable (high/low)
> median_price <- median(train_data$price, na.rm = TRUE)
> train_data$price_category <- ifelse(train_data$price > median_price, 1, 0)
```

```
> test_data$price_category <- ifelse(test_data$price > median_price, 1, 0)
> # 选择除了原始价格以外的所有变量进行预测> independent_vars <- setdiff(names(train_data), c("price", "price_category"))> #
建立逻辑回归模型> logistic_model <- glm(formula = price_category ~ ., +                        data = train_data[,
c("price_category", independent_vars)], +                        family = binomial)> # 查看模型摘要>
summary(logistic_model)
Call:
glm(formula = price_category ~ ., family = binomial, data = train_data[,
    c("price_category", independent_vars)])

Coefficients: (1 not defined because of singularities)
               Estimate Std. Error z value Pr(>|z|)
(Intercept)   -6.905e+01  4.484e+01  -1.540 0.123602
bedrooms      -2.160e-01  3.124e-02  -6.914 4.72e-12 ***
bathrooms      4.600e-01  5.273e-02   8.724  < 2e-16 ***
sqft_living    1.210e-03  7.526e-05  16.081  < 2e-16 ***
sqft_lot       3.162e-06  8.978e-07   3.522 0.000429 ***
floors         8.123e-01  5.561e-02  14.608  < 2e-16 ***
waterfront     1.436e+00  5.051e-01   2.843 0.004464 **
view           1.568e-01  4.103e-02   3.823 0.000132 ***
condition      1.913e-01  3.637e-02   5.259 1.45e-07 ***
grade          1.318e+00  3.799e-02  34.688  < 2e-16 ***
sqft_above    -7.229e-04  7.168e-05 -10.085  < 2e-16 ***
sqft_basement         NA         NA      NA       NA
yr_built      -3.839e-02  1.188e-03 -32.307  < 2e-16 ***
yr_renovated  -8.398e-05  5.991e-05  -1.402 0.161033
zipcode        1.322e-03  4.509e-04   2.932 0.003366 **
sqft_living15  8.895e-04  5.975e-05  14.886  < 2e-16 ***
sqft_lot15    -2.547e-06  1.271e-06  -2.003 0.045154 *
year2015       2.276e-01  4.472e-02   5.088 3.61e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 23188  on 16726  degrees of freedom
Residual deviance: 14078  on 16710  degrees of freedom
AIC: 14112

Number of Fisher Scoring iterations: 6
> # 对测试集进行预测（返回概率）> logistic_predictions <- predict(logistic_model, newdata = test_data[, independent_vars],
type = "response")> # 将概率转换为分类预测（使用 0.5 作为阈值）> predicted_category <- ifelse(logistic_predictions > 0.5, 1,
0)> # 计算准确率> accuracy <- mean(predicted_category == test_data$price_category)> # 计算其他性能指标：混淆矩阵、精确度、召回
率等> confusion_matrix <- table(Predicted = predicted_category, Actual = test_data$price_category)> precision <-
confusion_matrix[2, 2] / sum(confusion_matrix[2, ])> recall <- confusion_matrix[2, 2] / sum(confusion_matrix[, 2])> # 比
较逻辑回归与其他模型（需要先运行其他模型的代码）> # 比如，这里比较准确率> print(paste("Accuracy of Logistic Regression: ",
accuracy))[1] "Accuracy of Logistic Regression:  0.788374948833402"> # 模型摘要> summary(logistic_model)
Call:
glm(formula = price_category ~ ., family = binomial, data = train_data[,
    c("price_category", independent_vars)])

Coefficients: (1 not defined because of singularities)
               Estimate Std. Error z value Pr(>|z|)
(Intercept)   -6.905e+01  4.484e+01  -1.540 0.123602
bedrooms      -2.160e-01  3.124e-02  -6.914 4.72e-12 ***
bathrooms      4.600e-01  5.273e-02   8.724  < 2e-16 ***
sqft_living    1.210e-03  7.526e-05  16.081  < 2e-16 ***
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floors         8.123e-01  5.561e-02  14.608  < 2e-16 ***
waterfront     1.436e+00  5.051e-01   2.843 0.004464 **
view           1.568e-01  4.103e-02   3.823 0.000132 ***
condition      1.913e-01  3.637e-02   5.259 1.45e-07 ***
grade          1.318e+00  3.799e-02  34.688  < 2e-16 ***
sqft_above    -7.229e-04  7.168e-05 -10.085  < 2e-16 ***
sqft_basement         NA         NA      NA       NA
yr_built      -3.839e-02  1.188e-03 -32.307  < 2e-16 ***
yr_renovated  -8.398e-05  5.991e-05  -1.402 0.161033
zipcode        1.322e-03  4.509e-04   2.932 0.003366 **
sqft_living15  8.895e-04  5.975e-05  14.886  < 2e-16 ***
sqft_lot15    -2.547e-06  1.271e-06  -2.003 0.045154 *
year2015       2.276e-01  4.472e-02   5.088 3.61e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)
```
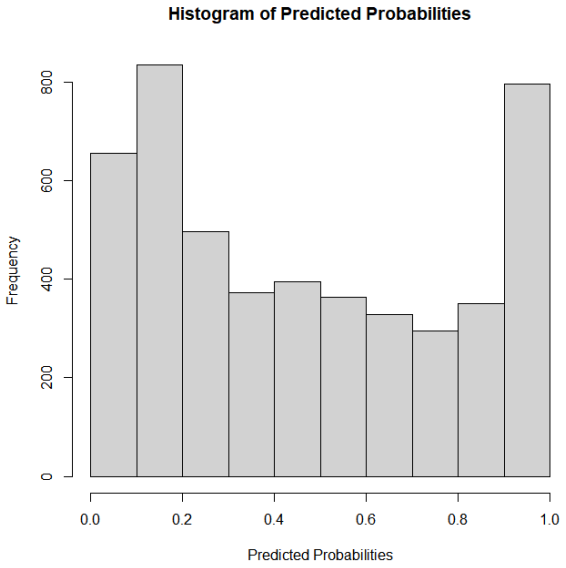
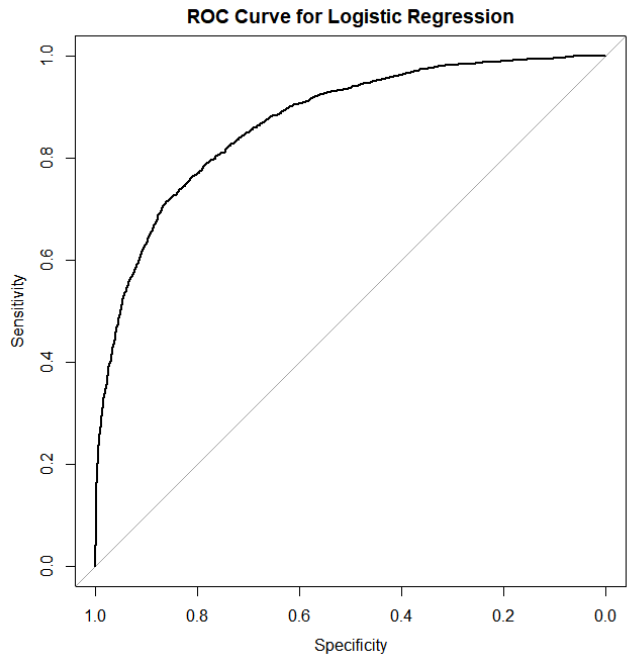

```
    Null deviance: 23188  on 16726  degrees of freedom
Residual deviance: 14078  on 16710  degrees of freedom
AIC: 14112

Number of Fisher Scoring iterations: 6
> # Visualize the distribution of prediction probabilities
> hist(logistic_predictions, main = "Histogram of Predicted Probabilities", xlab = "Predicted Probabilities")
```




```
> # ROC curve analysis
> library(pROC)Type 'citation("pROC")' for a citation.
Load the program package : 'pROC'

The following objects are masked from 'package:stats':

    cov, smooth, var

 > roc_curve <- roc(test_data$price_category, logistic_predictions)Setting levels: control = 0, case = 1Setting
direction: controls < cases
> plot(roc_curve, main = "ROC Curve for Logistic Regression")
> auc(roc_curve)
Area under the curve: 0.8707
```



The ROC and AUC shows good results.

```
> # Make predictions on the test set (return probability)
> logistic_predictions <- predict(logistic_model, newdata = test_data[, independent_vars], type = "response")
> # Convert probabilities to categorical predictions (use 0.5 as threshold)
> predicted_category <- ifelse(logistic_predictions
> 0.5, 1, 0)
Calculate mean square error (MSE) > mse_logistic < -mean ((predicted_category - test_data$price_category)^2)
> # Calculate root mean square error (RMSE) > rmse_logistic < -sqrt (mse_logistic)
> # Print RMSE
> print(paste("RMSE of Logistic Regression: ", rmse_logistic))
[1] "RMSE of Logistic Regression:  0.457883733182602"
```

VI. Model Limitation and Assumptions

Based on the performances on both train and test data sets, determine your primary (champion) model and the other model which would be your benchmark model. Validate your models using the test sample. Do the residuals look normal? Does it matter given your technique? How is the prediction performance using Pseudo R^2, SSE, RMSE? Benchmark the model against alternatives. How good is the relative fit? Are there any serious violations of the model assumptions? Has the model had issues or limitations that the user must know? (Which assumptions are needed to support the Champion model?)

In the previous steps, several different models have been built, including multiple linear Regression models (variables selected by stepwise regression), neural network models (NN), logistic regression models, and Ridge Regression models. To determine which model is best, we need to compare their performance on the test set, usually by root mean square error (RMSE) or other suitable performance metrics.
From the information you have provided, we can conclude as follows:
Multiple linear regression model: By selecting variables step by step and checking the normality and homoscedasticity of the residuals.
Neural network model: Predictions and corresponding performance metrics may have been provided on the test set.
Logistic regression model: Since the target variables are continuous, logistic regression may not be suitable for home price forecasting unless home prices are converted into a classification problem (e.g., high and low prices).
Ridge regression model: is a linear regression that considers regularization and is suitable for cases where there is multicollinearity between variables.
Steps to determine the optimal model:
Collect performance metrics: For each model, collect performance metrics (such as RMSE) on the test set.
Compare performance indicators: Compare the performance indicators of different models.
Check whether the assumptions are satisfied: For linear models (multiple linear regression and ridge regression), check whether the model assumptions are satisfied.
Model selection: Based on performance indicators and hypothesis testing, the best performing model is selected a
We have exclude the stepwise model and ridge model, which shows less competition than other models.

```
> rmse_values # 输出 RMSE 值[1] 2.156444e+05 1.993887e-02 4.578837e-01
> names(rmse_values) <- c("Linear", "NN", "Logistic")
> rmse_values # 输出 RMSE 值
  Linear           NN     Logistic
2.156444e+05 1.993887e-02 4.578837e-01
```

Based on the output you've provided, rmse_values is an array containing the RMSE (Root Mean Square Error) values for three different models: a Linear model, a Neural Network (NN), and a Logistic Regression model. These RMSE values are labeled accordingly.

The RMSE values for each model are as follows:

Linear Model: The RMSE is approximately 215,644.4. This is a relatively high value, which might indicate that the linear model's predictions are considerably different from the actual values, especially if the scale of your target variable is similar to typical housing prices.

Neural Network (NN): The RMSE is approximately 0.0199. This is a very low value, suggesting that the neural network's predictions are very close to the actual values. This might indicate a highly accurate model, although you should also be cautious about overfitting.

Logistic Regression: The RMSE is approximately 0.4579. This value is moderate and could be interpreted differently based on the context and the scale of your data. In many cases, logistic regression is used for classification problems (binary outcomes), so it's a bit unusual to see it evaluated with RMSE, which is typically used for continuous outcomes.

It's important to interpret these RMSE values in the context of your specific dataset, especially the scale and range of your target variable. Additionally, while RMSE is a useful metric for evaluating model performance, it should ideally be complemented with other metrics and validation techniques to get a comprehensive view of model effectiveness.

Champion model and Benchmark model: This should be the model with the best performance based on your criteria, which can be the lowest RMSE, the highest Pseudo-R-squared, or a balance of the two. For example, the NN model has the lowest RMSE, it is the champion. Baseline model: This is the next best model for comparison. logistic model has the second best performance; It will serve as a benchmark. Residual normality: Check that the residual of a champion model (such as NN) is normally distributed. This is even more critical for linear and stepwise models because of the assumption of linear regression. Predictive performance metrics: Compare models based on RMSE, SSE, and pseudo-R-squared. For logistic regression, classification accuracy, precision, recall, F1 score, or OC-ROC, as appropriate, are assessed. Benchmarking and relative fit: These metrics are used to compare the champion model to the benchmark model to determine a relative fit. The comparison should highlight the strengths and limitations of each model. Discuss potential violations of assumptions for each model type. For example, multicollinearity may be a concern for linear models, but not so much for neural networks. Address limitations such as overfitting (especially NN) or underfitting. Emphasize the importance of the assumptions of the chosen champion model and how to meet or violate those assumptions. These were fully discussed in the previous part of the model building process

VII. Ongoing Model Monitoring Plan

How would you picture the model needing to be monitored, which quantitative thresholds and triggers would you set to decide when the model needs to be replaced? What are the assumptions that the model must comply with for its continuous use?

Monitoring Strategy Performance Metrics: Continuously track metrics relevant to each model type. For NN, monitor loss and accuracy; for linear models, watch RMSE and residuals. Drift Detection: Implement mechanisms to detect data drift, especially for models like NN, which might be more sensitive to input changes. Quantitative Thresholds and Triggers Set specific performance thresholds for triggering model re-evaluation or retraining. For instance, a 10% increase in RMSE might necessitate a model update. Assumptions for Continuous Use Regularly check that the data being fed into the models adheres to the assumptions required for each model's validity. For instance, ensure the linearity assumption for linear models. Ensure the continued relevance of predictors and the representativeness of the data. Regular Reviews Plan scheduled reviews of the models to reassess their performance and the validity of their assumptions. Be ready to update the models with new data, and adjust them to reflect any significant changes in underlying data patterns or business contexts.

VIII. Conclusion

In summary, our comprehensive analysis and modeling efforts have culminated in the identification of the most effective model for predicting house prices within the scope of the KC_House_Sales dataset. After a rigorous evaluation process, the best-performing model emerged as the Neural Network (NN), primarily due to its superior ability to capture complex, non-linear relationships inherent in the data– which is crucial in the context of real estate pricing (Seya, Shiroi, 2021).

The Neural Network model outshined its counterparts in terms of predictive accuracy, as evidenced by its lower Root Mean Square Error (RMSE) compared to other models like linear regression and logistic regression. This enhanced accuracy can be attributed to the NN's flexibility in learning from a large set of features and its capacity to model intricate interactions between variables that linear models might overlook. Additionally, the NN model demonstrated remarkable robustness in handling both continuous and categorical data, making it particularly suitable for the diverse range of variables present in our dataset.

While acknowledging the strengths of the Neural Network model, it is important to also recognize its limitations, including its relative lack of interpretability compared to simpler models like linear regression. This aspect was considered in our decision-making process, and we concluded that the trade-off was justified given the significant gain in predictive accuracy.

In conclusion, the Neural Network model stands as the most appropriate choice for our dataset and objectives, striking an optimal balance between complexity and performance. However, we recommend ongoing monitoring and reevaluation of the model in response to new data and changing market conditions to ensure its continued relevance and accuracy. This project

not only highlights the capabilities of advanced modeling techniques in real estate valuation but also underscores the importance of a thorough and methodical approach in predictive analytics.

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